Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - II

Course Code: BCA104-2 **Max Marks: 100**

Course Name: DISCRETE MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Demonstrate a working knowledge of set notation and elementary set theory and recognize the connection between set operations and logic.

CO2: Apply the different properties of injections, surjections, bijections, compositions, and inverse functions.

CO3: Understand different types of matrices and operations among matrices.

CO4: Determine when a relation is reflexive, symmetric, antisymmetric, transitive, apply the properties of equivalence relations and partial orderings, and explain the connection between equivalence relations.

CO5: Effective express the elementary concepts and results of Number theory.

Section A **3*8=24**

1)	Discuss the properties of an equivalence relation on a set of integers using modular arithmetic.	**3**	(CO4, L2)
2) a)	Describe the properties that a relation must satisfy to be considered an equivalence relation. Give an example of a relation on the set {1, 2, 3} and classify it as reflexive, symmetric, antisymmetric, and transitive.	**3**	(CO4, L2)
	OR		
b)	Explain how to determine if a given relation on a set is antisymmetric using an example.	**3**	(CO4, L2)
3)	Illustrate with an example how to determine if a given relation on a set is both symmetric and transitive.	**3**	(CO1, L2)
4)	Examine and differentiate between an equivalence relation and a partial order on a given set A = {1, 2, 3, 4}. Provide examples for each and justify your answer by verifying the properties of reflexivity, symmetry, transitivity, antisymmetry, and totality.	**3**	(CO5, L4)
5) a)	Examine and differentiate between a symmetric matrix and an antisymmetric matrix using examples. Discuss their properties and provide one 3x3 example of each type.	**3**	(CO3, L4)
	OR		
b)	Examine and differentiate between a square matrix and an identity matrix by explaining their properties and providing an example for each.	**3**	(CO3, L4)

6)	Examine and differentiate between an equivalence matrix and a symmetric matrix, providing examples for each.	**3**	(CO3, L4)
7) a)	Discuss how to determine if a given relation on a set of integers is reflexive using an example.	**3**	(CO4, L2)
	OR		
b)	Discuss the properties of a partial ordering relation on a set A with an example to illustrate your answer.	**3**	(CO4, L2)
8)	Examine and differentiate between a finite set and an infinite countable set by providing examples for each and explaining their key properties.	**3**	(CO1, L4)

Section B **8*7=56**

9)	Consider an equivalence relation on the set = {a, b, c, d} represented by the matrix = [1 0 1 0; 0 1 0 1; 1 0 1 0; 0 1 0 1]. (i) Determine the elements of that are related to each other under . (ii) Construct a directed graph for the relation and show its equivalence classes. (iii) Compute the transitive closure of using Warshall's algorithm and verify if it remains an equivalence relation.	**8**	(CO3, L3)
10) a)	Consider the relation on the set = {0, 1, 2, 3, 4} defined by if and only if ≡ (2). Determine whether is reflexive, symmetric, antisymmetric, and transitive. Construct the matrix representation of and demonstrate its properties. Also, determine if is an equivalence relation or a partial ordering on .	**8**	(CO4, L3)
	OR		
b)	Determine whether the relation = {(,) ℕ × ℕ | + is even} is reflexive, symmetric, antisymmetric, or transitive. Provide proofs for your answers and explain how these properties relate to partial orderings if applicable.	**8**	(CO4, L3)
11)	Demonstrate through proof that the Cartesian product of two countable sets and is also a countable set. Provide a detailed explanation in your answer, including any relevant definitions or properties used.	**8**	(CO1, L3)
12) a)	Consider two matrices representing relations on the same set A = {1, 2, 3, 4} as follows: Matrix M_1 represents the relation R_1 and Matrix M_2 represents the relation R_2: M_1 = | 0 1 0 1 | | 1 0 1 0 | | 0 0 0 1 | | 0 1 0 0 | M_2 = | 0 1 0 0 | | 1 0 1 1 | | 0 0 0 1 | | 0 0 1 0 | (i) Determine the matrices for $R_1 \cap R_2$ and R_1 R_2. (ii) Analyze whether each of the resulting relations from part (i) is reflexive, symmetric, or transitive. Justify your answers with appropriate examples. (iii) Construct an example to show that the composition of R_1 and R_2 (R_1 R_2) does not necessarily commute.	**8**	(CO3, L4)
	OR		
b)	Analyze and compare the operations involving matrices with those in modular arithmetic. Specifically, explain the differences between matrix addition and subtraction versus addition and subtraction under modulo . Then, using a 3x3 matrix example, demonstrate both types of operations and discuss their applications in solving systems of linear equations.	**8**	(CO3, L4)

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13)	Determine whether the set of all integers divisible by 4 is countable or uncountable. If it is countable, construct a bijection between this set and the set of natural numbers. (i) Define what it means for a set to be countable. (ii) Prove that the set of all integers divisible by 4 is countable using the definition provided in part (i). (iii) Construct a function to demonstrate that there exists a one-to-one correspondence between the set of all integers divisible by 4 and the natural numbers.	**8**	(CO1, L3)
14) a)	Determine whether the set of all rational numbers between 0 and 1 (inclusive of 0 but exclusive of 1) is countable or uncountable, and provide a proof to support your answer.	**8**	(CO2, L3)
	OR		
b)	Let = {1, 2, 3, 4, 5} and consider two relations defined on as follows: Relation $_1$ = {(1, 2), (2, 3), (3, 4), (4, 5)} and Relation $_2$ = {(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (1, 3), (3, 5)}. (i) Determine whether each relation is injective, surjective, bijective. Justify your answer. (ii) Find the composition $_2$ $_1$ and represent it as a set of ordered pairs and then determine if it is symmetric or antisymmetric. (iii) Determine whether there exists an inverse function for each relation. If so, provide the inverse relations; otherwise, explain why no such inverse exists.	**8**	(CO2, L3)
15)	Explain how to determine if a given relation on the set of natural numbers = {1, 2, 3, ..., n} is a bijection. Provide examples to illustrate your explanation and prove whether the following relations are bijections or not: (i) R_1 = {(x, y) × | x + y ≤ 5}, (ii) R_2 = {(x, y) × | x - y is even}. Discuss the implications of your findings on the countability of these relations.	**8**	(CO2, L2)
	Section C		**10*2=20**
16)	Analyze and differentiate between a reflexive relation and an irreflexive relation using examples. Further, examine how these properties influence the formation of equivalence relations on the set A = {1, 2, 3, 4, 5}. Discuss whether any subset of A can form an equivalence class under each type of relation you have defined.	**10**	(CO4, L4)
17) a)	Discuss and explain the properties of injections, surjections, bijections, and their compositions in the context of proving that certain sets are countable. Specifically, for (i) an injection from set A to B, where |A| = n and |B| = m; (ii) a surjection from C onto D, with both sets having infinite elements; (iii) a bijection between set E and F, where E is the set of all positive integers and F is the set of all even positive integers. In each case, describe how these properties can be used to prove that a given set is countable.	**10**	(CO2, L2)
	OR		
b)	Discuss and illustrate with examples how a function can be classified as an injection, surjection, or bijection, using specific sets A = {1, 2, 3} and B = {4, 5, 6}. Additionally, explain the concept of composition of functions and determine whether the composite function is injective, surjective, or bijective when f: A → B and g: B → C are defined as follows: f(1) = 4, f(2) = 5, f(3) = 6; g(4) = 7, g(5) = 8, g(6) = 9.	**10**	(CO2, L2)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember

L2 - Understand

L3 - Apply

L4 - Analyse
L5 - Evaluate
L6 - Create